Muhammad Shoaib Asghar

Software Engineer

shoaibasghar1515@gmail.com +923186540421 Pakistan linkedin.com/in/mshoaibasghar

github.com/Shoaib1515

SUMMARY

Software Engineering graduate transitioning into cybersecurity, with hands-on experience building security-focused tools, backend APIs, and mobile UIs. Built SILI, a self-hosted AI cybersecurity assistant with vulnerability scanning and CVE-based network scanning, and a Secure Password Toolkit implementing PBKDF2-HMAC-SHA256, encrypted storage, and formal threat modeling. Comfortable with Python, FastAPI, and Flutter, with a growing focus on application security, secure coding practices, and vulnerability assessment. Quick learner who picks up new tools and domains fast and enjoys turning security requirements into working, well-documented software.

EXPERIENCE

Digisol 365 09/2024 – 10/2025
App Developer Bahawalpur

- Managed IT operations and support, including system maintenance, troubleshooting, network management, and software solutions.
- Contributed to project management, team coordination, and client support, demonstrating strong technical expertise and leadership.

EDUCATION

BuiltinSoft, Bahawalpur 07/2025 – 12/2025
Web Development Bootcamp

Excellence Delivered, Lahore 03/2025 – 06/2025
Advanced Python Bootcamp

Islamia University Bahawalpur 09/2020 – 05/2024
BS Software Engineering
CGPA: 3.34 / 4.00

TECHNICAL SKILLS

Languages
Python, JavaScript, SQL, C#, C++, HTML, CSS

Mobile Development
Flutter, responsive UI design, widget development

Backend & APIs
FastAPI, REST APIs, SQLAlchemy ORM, error handling, virtual environments

Databases
PostgreSQL, MySQL, MongoDB

Python Ecosystem
NumPy, Pandas, Jupyter Notebook, file handling, modules

Tools & OS
VS Code, PyCharm, Android Studio, Visual Studio, XAMPP, Anaconda Navigator, Kali Linux, Ubuntu

PROJECTS

SILI — AI Cybersecurity Assistant 06/2026 – 07/2026

- Built a self-hosted AI security tool from scratch that answers questions using live web search, scans code for vulnerabilities such as SQL injection and XSS, and runs authorized network scans using real CVE data.
- Integrated multiple APIs (Anthropic Claude, Groq, Serper, NVD) with a modular backend design so the underlying AI model can be swapped without touching the rest of the app.
- Implemented safety checks throughout — target authorization, injection-safe scanning, and rate limiting — to keep the tool legal and responsible to use.
- Designed and built the entire frontend independently, including a custom terminal-style UI.

Crime Investigation Management System 05/2026 – 06/2026

- Developed a backend REST API using FastAPI and PostgreSQL to manage detectives, cases, suspects, evidence, and witness statements.
- Applied relational database design and data validation, plus business rules such as case threat levels and AI-based risk scoring.

Secure Password Toolkit — Desktop Application 01/2026 – 02/2026

- Built a local-first credential manager in Python 3 (Tkinter) with authentication, encrypted storage, and a formal threat model and security assessment report.
- Implemented salted PBKDF2-HMAC-SHA256 key derivation (200,000 iterations) for login and per-entry encryption, keeping the session key in memory only.
- Enforced brute-force resistance (4-attempt lockout, 5-minute cooldown) and multi-user/multi-platform namespacing, with an optional Master Key for defense-in-depth.
- Added a real-time password strength meter and documented residual risks with hardening recommendations (e.g. migrating to AES-GCM).

Jarvis Voice Assistant 03/2025 – 05/2025

- Built a desktop AI assistant in Python for voice-controlled tasks — opening websites, reading the time aloud, running system commands, and playing music.
- Used speech_recognition, pyttsx3, and win32com.client to handle speech-to-text, text-to-speech, and system integration.

Social & Fintech UI Clones 09/2024 – 12/2024

- Developed front-end replicas of JazzCash, Facebook, and Instagram in Flutter, focusing on UI design, responsiveness, and user experience.
- Implemented interactive components and smooth navigation using widgets, rows, columns, containers, animations, and buttons.

Online Examination System — Final Year Project 01/2024 – 08/2024

- Built a web-based exam system where students log in with a university roll number, take exams, and receive instant results.
- Implemented a separate teacher portal for assigning and evaluating exams, using HTML, CSS, JavaScript, PHP, and MySQL (XAMPP).

SOFT SKILLS

- Problem Solving
- Technical Documentation
- Critical Thinking
- Time Management
- Team Collaboration
- Quick Learner
- Communication
- Adaptability